From Novel View Synthesis to Generative World Modeling: A Comprehensive Survey

ElephantFlow Research
github.com/elephantflow/ElephantSurvey
Draft v0.3 — March 2026

Abstract

Novel View Synthesis (NVS)—rendering a scene from arbitrary unseen viewpoints given reference observations—has undergone a paradigm shift from classical geometry-based reconstruction toward learning-driven generative approaches. This survey examines NVS not merely as a rendering technique but as the **foundational spatial capability** of Generative World Models: systems that predict visual scene states under arbitrary conditions including viewpoint changes, temporal dynamics, and agent interactions.

We present a unified conceptual framework organising existing methods along a *continuous spectrum*—from static multi-view reconstruction to dynamic, interactive world generation—across three axes: (i) scene representation (implicit, explicit, hybrid), (ii) generative paradigm (per-scene optimisation, feed-forward inference, diffusion-based synthesis), and (iii) world-modelling capability (geometry, dynamics, controllability). We survey more than **180 representative methods** spanning NeRF, 3D Gaussian Splatting, generalizable reconstruction, diffusion-based NVS, and 4-D world modeling, and propose a roadmap toward physically grounded, generalisable world models.

1 Introduction

1.1 The Fundamental Problem

Intelligence requires reasoning about the world beyond what is immediately visible. A self-driving vehicle must anticipate the scene around a blind corner; a robotic arm must predict how a grasped object will appear from the wrist camera after a planned motion; a surgeon must mentally reconstruct anatomy from a 2-D scan. In each case, the core operation is identical: **synthesising a plausible visual state at an unobserved vantage point**.

This is the problem of **Novel View Synthesis (NVS)**. Formally, given a set of reference observations $\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n$ with known camera poses $\boldsymbol{\pi}_i$, the goal is to render a photo-realistic image $\hat{\mathbf{I}}$ at an arbitrary target pose $\boldsymbol{\pi}^*$, faithfully preserving 3-D geometry, surface appearance, and lighting. The problem is inherently ill-posed when n is small or when the target pose is far from any reference: an infinite family of 3-D scenes is consistent with any finite set of 2-D projections. Resolving this ambiguity requires either dense capture (classical IBR) or a learned prior over the distribution of plausible scenes (modern neural and generative approaches).

While NVS was studied in the image-based rendering era [1–4], the past six years have witnessed an extraordinary paradigm shift: NeRF [17] demonstrated that a shallow MLP can represent complex scenes with view-dependent appearance by differentiably volume-rendering a continuous radiance field; 3DGS [46] showed that an explicit set of anisotropic Gaussians can render at real-time frame

Preprint.

Table 1: Comparison with related surveys. ✓ covered; ∼ partially; – not covered.

Survey	Year	NeRF	3DGS	Generative	Dynamic	Generalise	4D/WM	Unified
Tewari *et al.* [9]	2022	✓	–	∼	∼	∼	–	–
Gao *et al.* [10]	2022	✓	–	–	∼	–	–	–
Zhu *et al.* [11]	2023	∼	–	–	–	–	–	–
Wu *et al.* [12]	2024	–	✓	–	∼	–	–	–
Fei *et al.* [13]	2024	–	✓	∼	∼	–	–	–
Chen *et al.* [14]	2024	∼	∼	–	✓	–	∼	–
Zhang *et al.* [15]	2025	∼	∼	–	–	✓	–	–
He *et al.* [16]	2025	–	✓	∼	∼	–	–	–
Ours	2026	✓	✓	✓	✓	✓	✓	✓

rates; and diffusion-based methods [91, 102, 110] have moved the frontier toward *generalisable, generative* synthesis that requires as few as a single image—or even a text description—as input.

1.2 From NVS to World Modeling

We argue that this progression reflects a deeper conceptual shift: **NVS is becoming the foundational spatial inference engine of Generative World Modeling**.

> **Working Definition (World Model).** A *Generative World Model* is a learned system that can predict the visual (and optionally physical) state of a scene under arbitrary conditions—including viewpoint changes, temporal evolution, and agent interactions—in a manner that is *geometrically consistent*, *physically plausible*, and *generalisable* to scenes and conditions outside the training distribution.

Under this definition, NVS is the special case in which the only varying condition is camera pose. A full world model must additionally handle temporal dynamics (D3), action conditioning (D4), and physical plausibility (D5). Crucially, *spatial consistency enforced by NVS is a necessary prerequisite*: a world model without geometric grounding is merely a video predictor—perceptually plausible in distribution but unreliable for tasks requiring precise spatial reasoning such as robot manipulation or navigation.

Recent systems are actively closing this gap. Genie 2 [125] generates interactive 3-D environments from a single image; Cosmos [126] trains world foundation models at internet scale; and ViewCrafter [110] with StreetCrafter [200] demonstrate that camera-controlled video generation is practical at high resolution. These developments reflect the field's convergence on a unified model that subsumes NVS, video generation, and world simulation.

1.3 Relationship to Prior Surveys

Table 1 compares this work with the most closely related prior surveys. Tewari *et al.* [9] and Gao *et al.* [10] focus on NeRF representations without addressing world modelling. Wu *et al.* [12] and Fei *et al.* [13] cover 3DGS but not generative methods or 4-D. Chen *et al.* [14] cover NVS for autonomous driving but lack a world-modeling lens. To our knowledge, **no existing survey jointly covers all three paradigms (NeRF, 3DGS, generative) through the unifying lens of Generative World Modeling**, nor proposes a continuous-spectrum taxonomy from static reconstruction to 4-D interactive generation.

1.4 The Unified Spectrum

We organise all methods along a *continuous spectrum* defined by three axes (Figure 1):

Axis 1. Input regime: dense multi-view → sparse-view → single image → text/unconditional.

Axis 2. Temporal scope: static scene → deformable/dynamic → 4-D world generation.

Axis 3. Controllability: viewpoint only → viewpoint + time → viewpoint + time + action.

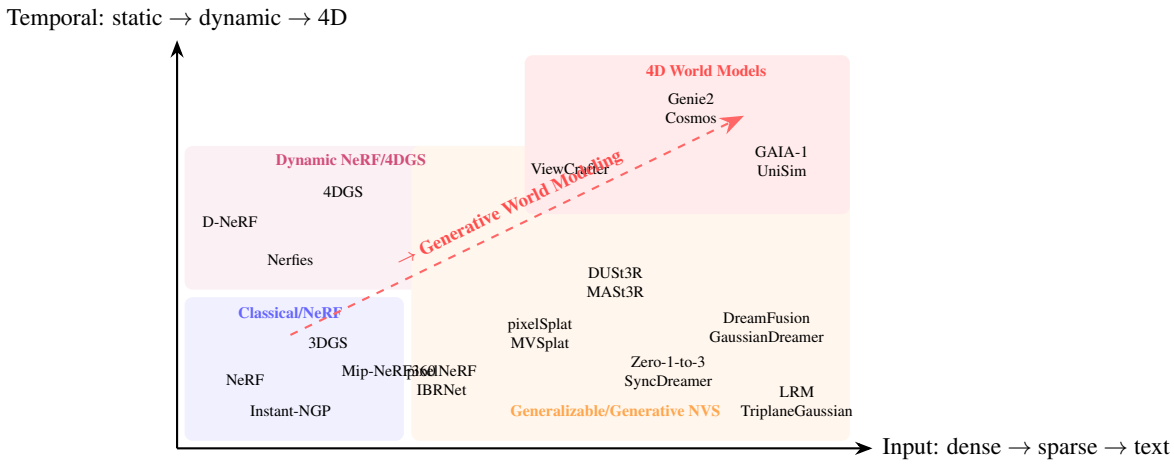

Figure 1: The unified NVS spectrum. The diagonal arrow marks the evolutionary trajectory toward 4-D Generative World Modeling.

1.5 Contributions and Paper Structure

Contributions: (1) A unified continuous-spectrum taxonomy across three axes (input regime, temporal scope, controllability); (2) Cross-paradigm coverage of >180 methods with explicit World Modeling implications in each chapter; (3) A related-survey gap analysis (Table 1) substantiating the novelty of the World Modeling lens; (4) A five-direction open-problems roadmap.

The remainder is organised as follows. §2 establishes background and a unified formulation. §4–6 survey the three representation paradigms (NeRF, 3DGS, feed-forward reconstruction). §7 covers generative NVS with particular depth on camera-controlled diffusion. §8 examines 4-D generation, neural driving simulators, and world foundation models. §9–10 review datasets and evaluation protocols. §11 proposes a research roadmap.

2 Background and Unified Formulation

2.1 Classical Image-Based Rendering

Before the neural era, NVS was approached through *image-based rendering* (IBR) [1–3]. The foundational insight is that images directly encode light transport: sufficiently dense sampling of the plenoptic function [6] permits view interpolation without reasoning about explicit 3-D geometry.

Light Field Rendering [1] and the Lumigraph [5] parameterise the light field on a two-plane slab, enabling photo-realistic view interpolation within the captured light-field volume. View interpolation [3] warps and blends reference images using depth estimates, anticipating modern depth-based view synthesis. Unstructured Lumigraphs and view-dependent texture mapping [4] relaxed capture requirements to unstructured camera sets. Structure from Motion (SfM) [7] and Multi-View Stereo (MVS) [8] established the geometry-estimation pipeline that underlies all pose-conditioned NVS to this day.

The fundamental limitation of classical IBR is the requirement for *dense* capture: interpolation quality degrades rapidly as target viewpoints move beyond the convex hull of the observed camera set. Furthermore, these methods produce no compact scene model—each novel view requires querying the full set of reference images at inference time. The desire for compact, generalisable scene representations drove the development of neural radiance fields.

2.2 Volume Rendering and NeRF

NeRF [17] represents a scene as a continuous function $F_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma)$, where $\mathbf{x} \in \mathbb{R}^3$ is a 3-D location and $\mathbf{d} \in \mathbb{S}^2$ a viewing direction. The expected colour of a ray $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$ is computed

via classical volume rendering [205]:

$$\hat{C}(\mathbf{r}) = \int_{s_n}^{s_f} T(s)\,\sigma(\mathbf{r}(s))\,\mathbf{c}(\mathbf{r}(s),\mathbf{d})\,\mathrm{d}s, \quad T(s) = \exp\left(-\int_{s_n}^{s} \sigma(\mathbf{r}(u))\,\mathrm{d}u\right), \tag{1}$$

where $T(s)$ is the accumulated transmittance. F_θ is a position-encoded MLP [206] trained by minimising $\|\hat{C}(\mathbf{r}) - C(\mathbf{r})\|_2^2$ over all observed rays. Hierarchical stratified sampling decomposes the continuous integral into coarse and fine networks, concentrating samples near the visible surface.

NeRF's key properties—implicit representation, continuous differentiability, and view-dependent appearance via directional encoding—made it transformative. Its central limitations are *slow per-scene optimisation* (hours) and *per-scene specificity* (no generalisation to unseen scenes), both of which the subsequent literature directly addressed.

2.3 Gaussian Splatting

3DGS [46] represents a scene as N anisotropic 3-D Gaussians $\{(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$, where $\boldsymbol{\mu}_i$ is the centre, $\boldsymbol{\Sigma}_i$ the full covariance, α_i the opacity, and $\mathbf{c}_i$ a set of spherical-harmonic coefficients encoding view-dependent colour. Novel views are rendered by projecting each Gaussian to a 2-D splat and compositing front-to-back:

$$\hat{C} = \sum_{i \in \mathcal{N}} \mathbf{c}_i\,\alpha_i\,\tilde{G}_i \prod_{j<i}(1 - \alpha_j\,\tilde{G}_j), \tag{2}$$

where $\tilde{G}_i$ denotes the 2-D projection of Gaussian i. Training proceeds by differentiable rasterisation and minimisation of a combined ℓ_1 and SSIM photometric loss. An *adaptive densification* strategy clones under-reconstructed Gaussians and splits over-large ones, while pruning transparent or redundant primitives.

3DGS achieves real-time rendering (>100 fps at 1080p) with training times of tens of minutes—a dramatic improvement over NeRF. Its explicit, editable representation makes it attractive for downstream tasks in scene editing, simulation, and embodied AI. The primary limitations are *higher memory footprint* (millions of Gaussians per scene) and *limited surface quality* compared to SDF-based methods. A large subsequent literature has addressed both limitations.

2.4 Unified NVS and World Modeling Formulation

We formulate NVS as a general function:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n,\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}^*, \quad n \in [1, +\infty). \tag{3}$$

The world-modeling extension augments this with a time index t and a sequence of agent actions $\mathbf{a}_{0:t}$:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n,\ \mathbf{a}_{0:t},\ t,\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}_t^*. \tag{4}$$

Eq. (3) is the special case $t = 0$, $\mathbf{a} = \varnothing$. A complete world model satisfying Eq. (4) must fulfil five desiderata, which we use throughout the survey: **D1** geometric consistency across viewpoints; **D2** generalisation from sparse or single-image input; **D3** dynamic scene modelling; **D4** action conditioning; **D5** physical plausibility.

3 Scene Representation Taxonomy

The choice of scene representation determines the fundamental trade-offs a method faces: rendering speed, reconstruction accuracy, memory footprint, generalisability, and editability. We identify four families, each reflecting a different position on the speed–compactness–quality Pareto frontier.

Implicit representations (MLPs, hash grids) encode the scene as a continuous function $f_\theta(\mathbf{x})$. Their key advantages are compactness and continuous differentiability, which support high-quality reconstruction from dense observations. Their central limitation is rendering speed: each pixel requires integrating along a ray through many network evaluations. NeRF [17] and its successors (§4) are the canonical family.

Explicit representations (voxels, point clouds, Gaussians) store scene attributes at discrete spatial elements, enabling tile-based or splat-based rasterisation that achieves real-time frame rates. The

trade-off is memory: a faithful 3DGS scene may occupy hundreds of megabytes. 3D Gaussian Splatting [46] (§5) is the dominant current representative.

Hybrid representations combine an explicit spatial structure—typically a triplane [88] or multi-resolution grid [23]—with a shallow MLP decoder. They strike a practical balance between the quality of implicit models and the speed of explicit ones, and are the backbone of most large reconstruction models (§6).

Compositional representations decompose the scene into semantic entities (objects, backgrounds, agents) with independent sub-representations. This structural inductive bias enables object-level editing and compositional generation [59, 87], but typically requires supervision or segmentation to separate entities. It is also the natural representation for world models that must reason about independently acting objects.

4 NeRF: Implicit Neural Radiance Fields

4.1 Foundational NeRF Methods

The original NeRF [17] demonstrated that an MLP with positional encoding could faithfully represent complex real scenes with view-dependent appearance from a few dozen posed photographs. Subsequent work systematically addressed its two principal limitations—slow optimisation and poor generalisation to novel scenes.

Accelerating per-scene optimisation. The bottleneck in vanilla NeRF is the MLP query: every rendered pixel requires hundreds of network evaluations along the ray. Instant-NGP [23] replaces the large MLP backbone with a compact multi-resolution *hash grid*, enabling training convergence in under a minute on a single GPU while matching or exceeding prior quality. TensoRF [24] decomposes the radiance field into low-rank CP and VM tensor components, reducing both storage and rendering overhead. Plenoxels [25] eliminates the MLP entirely, using a sparse 3-D voxel grid with trilinearly interpolated spherical harmonics; the resulting model trains in minutes without gradient-intensive network backward passes. DirectVoxGO [26] accelerates convergence via a two-stage coarse-to-fine voxel grid that aggressively skips empty space.

Improving visual quality and handling challenging scenes. Mip-NeRF [19] replaces the point-sampled ray with an *integrated positional encoding* over a conical frustum, substantially reducing aliasing at multiple viewing distances. Mip-NeRF 360 [20] extends this to unbounded outdoor scenes via a non-linear scene parameterisation that contracts the infinite background into a bounded representation. Zip-NeRF [21] unifies the hash grid of Instant-NGP with the anti-aliasing of Mip-NeRF 360, establishing the current quality frontier for per-scene NeRF. NeRF-W [18] introduces per-image latent appearance codes and a transient object model, enabling reconstruction of tourist-photographed landmarks from uncontrolled internet images. Ref-NeRF [22] restructures the directional encoding into a physically-based reflection model that explicitly predicts diffuse colour, specular tint, and surface roughness, dramatically improving the rendering of specular and metallic materials.

Surface reconstruction. While volume rendering naturally produces accurate alpha composites, it does not enforce a well-defined surface, leading to noisy geometry. NeuS [27] reformulates volume rendering using a logistic density derived from a Signed Distance Function (SDF), yielding watertight, high-quality surface reconstruction. VolSDF [28] derives a rigorous analytical mapping from SDF values to volume density, providing a theoretically grounded alternative. BakedSDF [29] combines the surface quality of SDF-based NeRF with real-time rendering by extracting and baking the learned SDF into a high-quality mesh.

Table 2 provides a comprehensive comparison of NeRF variants across all three sub-families: per-scene optimisation, generalizable, and dynamic methods.

4.2 Generalizable NeRF

Per-scene NeRF requires lengthy per-scene optimisation, limiting scalability. *Generalisable* methods learn a shared prior across scenes, enabling fast inference on new scenes.

Table 2: Comprehensive NeRF method comparison. **Category**: PS=per-scene, GEN=generalizable, DYN=dynamic. **Train/Render**: speed relative to vanilla NeRF (Fast/Med/Slow). **Gen.**: generalizes across scenes without per-scene re-training. **Dyn.**: handles dynamic or deformable scenes. **Surf.**: produces explicit surface (mesh/SDF).

Method	Venue	Cat.	Train	Render	Gen.	Dyn.	Surf.	Key Contribution
Per-scene optimisation								
NeRF [17]	ECCV'20	PS	Slow	Slow	–	–	–	MLP + positional encoding
NeRF-W [18]	CVPR'21	PS	Slow	Slow	–	–	–	In-the-wild appearance codes
Mip-NeRF [19]	ICCV'21	PS	Slow	Slow	–	–	–	Integrated positional encoding
NeuS [27]	NeurIPS'21	PS	Slow	Slow	–	–	✓	SDF-based volume rendering
VolSDF [28]	NeurIPS'21	PS	Slow	Slow	–	–	✓	Analytical SDF-to-density
Plenoxels [25]	CVPR'22	PS	Fast	Fast	–	–	–	No-MLP voxel grid + SH
TensoRF [24]	ECCV'22	PS	Med	Fast	–	–	–	CP/VM tensor decomposition
DirectVoxGO [26]	CVPR'22	PS	Fast	Fast	–	–	–	Two-stage coarse-to-fine voxel
Instant-NGP [23]	TOG'22	PS	Fast	Fast	–	–	–	Multi-resolution hash grid
Mip-NeRF360 [20]	CVPR'22	PS	Slow	Slow	–	–	–	Unbounded scene contraction
Ref-NeRF [22]	CVPR'22	PS	Slow	Slow	–	–	–	Physics-based reflection model
Zip-NeRF [21]	ICCV'23	PS	Fast	Fast	–	–	–	Hash grid + anti-aliasing
BakedSDF [29]	SIGGRAPH'23	PS	Med	Fast	–	–	✓	SDF mesh extraction + baking
Generalizable methods								
pixelNeRF [30]	CVPR'21	GEN	Fast	Slow	✓	–	–	Pixel-aligned CNN features
IBRNet [31]	CVPR'21	GEN	Fast	Slow	✓	–	–	Cross-view transformer aggregation
MVSNeRF [32]	ICCV'21	GEN	Fast	Slow	✓	–	–	Cost-volume geometry prior
SRT [34]	CVPR'22	GEN	Fast	Fast	✓	–	–	Set-latent scene representation
GNT [33]	arXiv'22	GEN	Fast	Med	✓	–	–	Fully attention-based, no NeRF MLP
NerfDiff [35]	ICML'23	GEN	Fast	Slow	✓	–	–	Diffusion prior for single-view
ZeroRF [36]	arXiv'23	GEN	Med	Slow	∼	–	–	Zero-pretrain sparse-view 360°
FrameNeRF [162]	arXiv'24	GEN	Fast	Slow	✓	–	–	Video frames as pseudo views
Dynamic & deformable methods								
D-NeRF [37]	CVPR'21	DYN	Slow	Slow	–	✓	–	Canonical deformation network
NSFF [38]	CVPR'21	DYN	Slow	Slow	–	✓	–	Explicit scene flow field
Nerfies [39]	ICCV'21	DYN	Slow	Slow	–	✓	–	Per-frame latent deformation
HyperNeRF [40]	TOG'21	DYN	Slow	Slow	–	✓	–	Higher-dimensional ambient space
Non-Rigid NeRF [196]	ICCV'21	DYN	Slow	Slow	–	✓	–	Monocular ray bending
DyNeRF [41]	CVPR'22	DYN	Slow	Slow	–	✓	–	Time-conditioned multi-view NeRF
TiNeuVox [42]	SIGGRAPH'22	DYN	Med	Med	–	✓	–	Voxel-accelerated dynamic NeRF
K-Planes [43]	CVPR'23	DYN	Fast	Fast	–	✓	–	4-D plane decomposition
HexPlane [44]	CVPR'23	DYN	Fast	Fast	–	✓	–	Concurrent 4-D plane factorisation
NeRFPlayer [45]	TOG'23	DYN	Med	Med	–	✓	–	Static/deforming/new region decomp.
Soup-of-Planes [183]	ICCV'23	DYN	Fast	Fast	–	✓	–	Planar fitting for casual video
Pseudo-GDV [184]	CVPR'24	DYN	Med	Slow	–	✓	–	Static + dynamic inpainting

pixelNeRF [30] conditions the NeRF MLP on pixel-aligned CNN features extracted from reference views, enabling single-image reconstruction. IBRNet [31] uses a transformer to aggregate features from multiple reference views at query points along the ray. MVSNeRF [32] builds a cost volume from stereo for geometry-aware feature extraction. GNT [33] dispenses with the NeRF MLP entirely, replacing it with a two-stage transformer that first encodes reference views into a view-independent representation and then renders the target view via cross-attention, learning geometric reasoning as an emergent property of attention rather than an explicit inductive bias. SRT [34] pursues a similar philosophy but uses a set-latent architecture, encoding an arbitrary number of views into a fixed-size scene representation without pose assumptions. NerfDiff [35] augments generalizable NeRF with diffusion priors for single-image synthesis. ZeroRF [36] achieves sparse-view 360° reconstruction via per-scene optimisation with zero pretraining, leveraging factorised feature grids.

Few-shot and test-time methods. FrameNeRF [162] presents a simple and efficient framework for few-shot NVS by exploiting temporal video frames as dense pseudo views. Recollection from Pensieve [173] learns NVS from uncalibrated internet videos without posed supervision. Test-Time Completion [195] repurposes natural video completion as a zero-shot NVS prior at test time. Mixed Neural Voxels [174] enables fast multi-view video synthesis by decomposing scenes into static and dynamic voxel grids.

4.3 Dynamic NeRF

Modelling non-rigid dynamic scenes is a central challenge for world modeling.

D-NeRF [37] introduces a deformation network mapping each spacetime point to a canonical static space. Nerfies [39] and HyperNeRF [40] model complex non-rigid deformations via latent codes and higher-dimensional ambient spaces. Neural Scene Flow Fields (NSFF) [38] models dynamic scenes with scene flow representation. Non-Rigid NeRF [196] reconstructs and synthesises novel views of dynamic scenes from monocular video with explicit ray bending. DyNeRF [41] extends NeRF to monocular video via a time-conditioned model. TiNeuVox [42] uses voxel features to accelerate dynamic NeRF. K-Planes [43] decomposes 4-D space-time into six feature planes—three spatial (XY, XZ, YZ) and three spatial-temporal (XT, YT, ZT)—each encoded as a learned 2-D grid, making the architecture highly memory-efficient and interpretable. HexPlane [44] independently proposes the same factorisation, demonstrating its generality; together these works establish plane-based 4-D decomposition as the dominant acceleration paradigm for dynamic NeRF. NeRFPlayer [45] separately encodes static, deforming and newly appearing regions. Soup-of-Planes [183] enables fast view synthesis of casual monocular videos by fitting a set of planar scenes. Pseudo-Generalised Dynamic View [184] synthesises dynamic views from monocular video by combining static and dynamic video inpainting. Decoupled Dynamic Video [179] disentangles foreground dynamics and background statics for monocular dynamic view synthesis.

The dynamic NeRF methods are included in the comprehensive Table 2 above.

5 3D Gaussian Splatting

5.1 Foundational 3DGS

3DGS [46] achieved real-time ($>$100 fps at 1080p) NVS with training times of 30–45 minutes on commodity hardware, surpassing NeRF-quality baselines on standard benchmarks. Its three key innovations are: (i) anisotropic Gaussian primitives that align naturally with planar scene structures; (ii) tile-based rasterisation that enables massively parallel GPU rendering; (iii) adaptive densification that automatically allocates capacity to under-reconstructed regions. The resulting representation is fully explicit and editable, opening new directions in scene editing, simulation, and avatar rendering.

Quality and anti-aliasing. Mip-Splatting [47] identifies aliasing in vanilla 3DGS as arising from the approximation of 3-D Gaussian convolution during 2-D projection. It introduces a 3-D smoothing filter derived from the sampling footprint, enabling accurate multi-scale rendering and substantially reducing floaters at large distances. AbsGS [48] revisits the densification criterion: replacing the mean gradient with an absolute-value measure improves coverage in thin structures and reduces needle-like artifacts near object boundaries.

Compression and memory efficiency. A single 3DGS scene may require 300MB–1GB of storage, limiting deployment. Scaffold-GS [49] introduces a sparse set of anchor primitives from which neural networks predict neighbouring Gaussian attributes; this structured representation reduces storage by 3–5$\times$ with comparable quality. LightGaussian [50] achieves 15$\times$ compression via global Gaussian pruning based on visibility contribution and subsequent vector quantisation of attributes. EAGLES [52] applies residual vector quantisation for fine-grained attribute compression. HAC [53] proposes a hash-grid-based context model that predicts per-Gaussian attribute entropy, enabling learned arithmetic coding for extreme compression ratios.

Surface reconstruction. SuGaR [54] aligns Gaussians to the scene surface by adding a regularisation term that encourages Gaussians to become flat discs at the surface, followed by mesh extraction and joint Gaussian–mesh refinement. 2DGS [55] replaces 3-D Gaussians with 2-D disc primitives, which by construction lie on surfaces and produce accurate normal maps. Gaussian Opacity Fields (GOF) [56] fuses Gaussian Splatting with a continuous opacity field for topologically correct surface extraction in unbounded scenes.

Scene editing and compositionality. The explicit Gaussian representation is particularly amenable to editing: individual primitives can be identified, moved, duplicated, or removed. GaussianEditor [57] harnesses a large vision-language model to select semantically relevant Gaussians from a text description and optimises their appearance with a diffusion prior. Gaussian Grouping [58] augments each Gaussian with a learned identity field, trained with SAM [211] masks from rendered views,

Table 3: Comprehensive 3D Gaussian Splatting method comparison. **Category**: STATIC=per-scene static, DYN=dynamic/deformable, FF=feed-forward generalizable. **RT**: real-time rendering (>30 fps). **Comp.**: dedicated compression or storage reduction. **Surf.**: explicit surface extraction. **Unp.**: works on unposed images. **Input**: minimum input views for FF methods.

Method	Venue	Cat.	RT	Comp.	Surf.	Edit.	Unp.	Input	Key Contribution
Static per-scene									
3DGS [46]	TOG'23	S	✓	–	–	–	–	MV	Tile-based Gaussian splatting
Mip-Splatting [47]	CVPR'24	S	✓	–	–	–	–	MV	3-D smoothing filter, anti-alias
AbsGS [48]	NeurIPS'24	S	✓	–	–	–	–	MV	Absolute-gradient densification
Scaffold-GS [49]	CVPR'24	S	✓	✓	–	–	–	MV	Anchor-driven neural attributes
LightGaussian [50]	NeurIPS'23	S	✓	✓	–	–	–	MV	Visibility pruning + VQ, 15×
Compact3D [51]	arXiv'24	S	✓	✓	–	–	–	MV	k-means codebook compression
EAGLES [52]	ECCV'24	S	✓	✓	–	–	–	MV	Residual vector quantisation
HAC [53]	ECCV'24	S	✓	✓	–	–	–	MV	Hash context entropy coding
SuGaR [54]	CVPR'24	S	✓	–	✓	∼	–	MV	Flat-disc + mesh joint opt.
2DGS [55]	SIGGRAPH'24	S	✓	–	✓	–	–	MV	2-D disc primitives
GOF [56]	arXiv'24	S	✓	–	✓	–	–	MV	Opacity field for surface
GaussianEditor [57]	CVPR'24	S	✓	–	–	✓	–	MV	LLM-guided semantic editing
Gaussian Grouping [58]	ECCV'24	S	✓	–	–	✓	–	MV	SAM-based identity field
GALA3D [59]	ICML'24	S	✓	–	–	✓	–	MV	LLM-layout compositional gen.
LERF [135]	ICCV'23	S	–	–	–	∼	–	MV	CLIP language grounding in NeRF
LangSplat [136]	CVPR'24	S	✓	–	–	∼	–	MV	Language field in 3DGS
Dynamic & deformable									
4DGS [60]	CVPR'24	D	✓	–	–	–	–	MV	4-D Gaussian + temporal field
Deformable3DGS [62]	ICLR'24	D	✓	–	–	–	–	Mono	Canonical deformation network
SpacetimeGaussians [61]	CVPR'24	D	✓	–	–	–	–	MV	Polynomial time trajectories
SC-GS [63]	CVPR'24	D	✓	–	–	–	–	Mono	Sparse control point deform.
Gaussian-Flow [64]	CVPR'24	D	✓	–	–	–	–	MV	Fourier polynomial motion
Shape-of-Motion [65]	arXiv'24	D	–	–	–	–	–	Mono	Per-Gaussian 3-D motion
Dynamic 3DGS [175]	3DV'24	D	✓	–	–	–	–	MV	Persistent Gaussian tracking
GauFRe [191]	arXiv'24	D	✓	–	–	–	–	Mono	Flow-regularised deformation
MoVieS [192]	arXiv'24	D	✓	–	–	–	–	Mono	4-D synthesis in ∼1 second
SC-4D [176]	arXiv'24	D	–	–	–	–	–	Mono	Self-calibrating 4-D from video
Reconstruct-Inpaint [177]	arXiv'24	D	–	–	–	–	–	Mono	Recon + inpaint + TTF pipeline
Articulated-GS [194]	NeurIPS'24	D	✓	–	–	–	–	MV	Template-free articulated 3DGS
EVA-Gaussian [185]	arXiv'24	D	✓	–	–	–	–	MV	Real-time human NVS
Feed-forward generalizable									
pixelSplat [66]	CVPR'24	FF	✓	–	–	–	–	2	Stereo epipolar features
MVSplat [67]	ECCV'24	FF	✓	–	–	–	–	2–6	Cost-volume geometry prior
TranSplat [68]	arXiv'24	FF	✓	–	–	–	–	2+	Transformer MV prediction
FSGS [69]	CVPR'24	FF	✓	–	–	–	–	3	Gaussian unpooling, few-shot
DepthSplat [70]	arXiv'24	FF	✓	–	–	–	–	2+	Joint depth + Gaussian pred.
Splatt3R [71]	arXiv'24	FF	✓	–	–	–	✓	2	DUSt3R backbone, unposed
TriplaneGaussian [72]	CVPR'24	FF	✓	–	–	–	–	1	Triplane + Gaussian, single-view

enabling instance-level part decomposition and editing. GALA3D [59] uses an LLM to generate a 3-D layout from text and initialises Gaussians per-object, enabling compositional scene generation.

Table 3 provides a comprehensive view of the 3DGS family.

5.2 Dynamic 3DGS

4DGS [60] extends Gaussians into 4-D space-time with a temporal radiance field. SpacetimeGaussians [61] assigns polynomial time trajectories to each Gaussian. Deformable3DGS [62] applies a deformation network to canonical Gaussians. SC-GS [63] uses sparse control points for monocular dynamic scenes. Gaussian-Flow [64] models Gaussian dynamics via polynomial motion fields in Fourier space. Shape-of-Motion [65] reconstructs per-Gaussian 3-D motion from monocular videos. Dynamic 3D Gaussians [175] tracks scene dynamics through persistent Gaussian views, jointly optimising NVS and tracking. GauFRe [191] uses Gaussian deformation fields with flow regularisation for real-time dynamic NVS. MoVieS [192] generates motion-aware 4-D dynamic views in approximately one second via efficient Gaussian motion estimation. Self-Calibrating 4D [176] performs 4-D NVS from monocular videos by jointly estimating camera parameters and Gaussian deformations. Reconstruct-Inpaint-Finetune [177] addresses dynamic monocular NVS via a three-stage pipeline of reconstruction, inpainting, and test-time refinement. EVA-Gaussian [185] achieves real-time human NVS under diverse multi-view settings. Articulated Gaussian [194] enables template-free articulated Gaussian Splatting for real-time reposable dynamic view synthesis.

5.3 Feed-Forward Generalizable 3DGS

pixelSplat [66] predicts Gaussian primitives from stereo image pairs in a single forward pass. MVSplat [67] adds cost-volume priors for better geometry. TranSplat [68] introduces a transformer architecture for multi-view Gaussian prediction. Splatt3R [71] builds on DUSt3R [73] for unposed image pairs. FSGS [69] enables few-shot Gaussian Splatting via Gaussian unpooling and monocular depth supervision. DepthSplat [70] jointly predicts depth and Gaussians for generalizable NVS. TriplaneGaussian [72] combines triplane and Gaussian representations in a transformer-based single-view reconstruction model.

The feed-forward generalizable methods are included in the comprehensive Table 3 above (FF category).

6 Generalisable Feed-Forward 3D Reconstruction

6.1 Pose-Free and Foundation Models

A new generation of methods aims for *pose-free* generalizable reconstruction. DUSt3R [73] predicts dense pointmaps from arbitrary image pairs without camera calibration, then aligns them via global optimisation. MASt3R [74] adds matching capabilities to DUSt3R, enabling accurate localisation. MonST3R [75] extends DUSt3R to dynamic video. VGGT [76] is a visual geometry grounded deep structured tokens model for feed-forward 3-D structure prediction.

6.2 Large Reconstruction Models (LRM)

The Large Reconstruction Model (LRM) [77] paradigm represents a qualitative shift from per-scene optimisation to *feed-forward inference*: a single forward pass of a large transformer, pretrained on hundreds of thousands of 3-D assets, produces a NeRF or Gaussian scene from a single image in under 5 seconds. This is possible because the model amortises the geometric reasoning of NeRF optimisation into its weights—a striking demonstration that scale can substitute for per-instance compute.

LRM [77] itself uses an image encoder (DINO-ViT) followed by a cross-attention transformer that maps image tokens to triplane tokens, which are then decoded by a NeRF MLP. Instant3D [78] extends this by first generating sparse multi-view images with a diffusion model, then feeding the result to an LRM-style reconstructor, decoupling appearance generation from geometry prediction. OpenLRM [79] reproduces LRM in an open-source setting.

The Gaussian-based branch has proven especially fast. LGM [80] generates multi-view images and lifts them to a 3DGS scene via an asymmetric U-Net, achieving 3-D generation in seconds at commercial-grade quality. GRM [244] predicts 3D Gaussians directly from image features using a sparse-view transformer, operating entirely at the pixel level without any intermediate NeRF. Gamba [245] replaces the transformer with a state-space model (Mamba) for $O(N)$ sequence length scaling, enabling long-context 3-D inference. GS-LRM [246] combines a transformer backbone with a 3DGS output head, achieving state-of-the-art single-image reconstruction quality.

Two non-parametric reconstruction models have recently emerged as strong baselines. DUSt3R [73] formulates 3-D reconstruction as point-map regression: given an image pair, a ViT predicts dense 3-D coordinates for every pixel in a shared Euclidean frame, requiring no camera poses at test time. MASt3R [74] extends DUSt3R to feature-level matching, enabling accurate metric reconstruction from internet-sourced photo pairs. These models underpin a new generation of unposed feed-forward reconstructors such as Splatt3R [71], which produces Gaussian splatting scenes directly from uncalibrated image pairs.

Implications for World Modeling. LRM-style models demonstrate that a single transformer, with sufficient data and compute, can learn a universal spatial prior—the same capacity required by world models for instant scene initialisation from single-frame observations.

6.3 Multi-View Stereo in the Neural Era

Classical MVS has been reinvigorated by deep learning. MVSNet [82] introduced cost-volume-based deep MVS. IterMVS [83] and Uni-MVSNet [84] improved accuracy and generalisation. These methods provide the geometry backbone for downstream NVS, and their cost-volume feature matching is directly incorporated into generalizable NeRF (MVSNeRF [32]) and feed-forward Gaussian methods (MVSplat [67]).

7 Generative Paradigms for Novel View Synthesis

The emergence of deep generative models—first GANs and then diffusion models—has fundamentally altered the landscape of NVS. Where reconstruction-based methods infer a scene representation from observations and render it deterministically, *generative* methods learn a distribution over plausible visual completions given partial observations. This distinction is not merely architectural: it represents a shift from *interpolation* (rendering within the observed convex hull of viewpoints) to *extrapolation* (hallucinating plausible content beyond it). Generative methods are therefore the primary vehicle for the one-image-to-3D, text-to-3D, and single-image world-generation capabilities that bring NVS into the world-modeling regime.

7.1 3-D-Aware GANs: Learning Geometry from 2-D Supervision

The first wave of generative NVS methods embedded radiance fields inside GAN generators, enabling implicit 3-D supervision from unposed 2-D image collections—a paradigm that proved influential precisely because it required no 3-D ground truth.

GRAF [85] pioneered this direction by conditioning a NeRF generator on a latent shape-appearance code, training with a 2-D discriminator. pi-GAN [86] improved the NeRF backbone with periodic activations (SIREN [210]), achieving substantially better multi-view consistency. GIRAFFE [87] introduced scene compositionality by combining foreground and background neural feature fields, enabling object-level editing without explicit segmentation. EG3D [88] is the landmark method of this era: its *triplane* representation—three axis-aligned 2-D feature maps decoded by a shallow MLP—reduces rendering cost by orders of magnitude and admits a super-resolution upsampler, enabling 512×512 training at practical throughput. StyleNeRF [89] integrates the NeRF volume renderer into a StyleGAN [209] backbone, inheriting its powerful style-based latent structure for fine-grained appearance control.

A critical limitation of GAN-based approaches is *mode coverage*: the discriminator objective encourages sharp, realistic individual samples but does not explicitly minimise distributional divergence over all plausible views, making it prone to mode dropping for complex, asymmetric scenes. This motivated the transition toward diffusion-based approaches.

7.2 Diffusion Models for Camera-Controlled NVS

Diffusion models [90, 207] have become the dominant generative paradigm for NVS due to three decisive advantages: (i) training stability via a simple denoising objective; (ii) access to powerful image priors from large-scale pretraining; (iii) natural integration of arbitrary conditioning signals (pose, depth, text, reference image) through classifier-free guidance.

7.2.1 Pose-Conditioned Single-Image NVS

The most direct application is *single-image NVS*: given one reference photograph and a target camera pose π^*, generate a plausible rendering at that viewpoint. This is under-constrained without a prior—an infinite number of 3-D scenes are consistent with a single 2-D observation—and diffusion models provide exactly the distributional prior needed.

Zero-1-to-3 and successors. Zero-1-to-3 [91] established the canonical formulation: fine-tune a latent diffusion model (Stable Diffusion [90]) on multi-view renders from Objaverse [146], conditioning on both the reference image and the relative pose $(\Delta\phi, \Delta\theta, \Delta r)$ encoded as a CLIP-space vector. The model learns to "imagine" how an object would look from any viewpoint, without constructing an explicit 3-D representation. Zero123++ [94] strengthens multi-view consistency by

predicting a fixed grid of six canonical views simultaneously, reducing inter-view inconsistency from independent sampling. One-2-3-45 [93] pipelines Zero-1-to-3 outputs into a cost-volume-based MVS reconstruction [82], converting the diffusion output to a textured mesh within 45 seconds.

Synchronised multi-view denoising. A key challenge with independently conditioned diffusion is that each generated view is sampled from a different realisation of the latent distribution, producing inter-view geometric contradictions. SyncDreamer [92] addresses this by coupling the denoising of multiple target views through shared 3-D volume attention at each diffusion step, enforcing consistent geometry throughout the synthesis process. MVDiffusion [95] constructs a correspondence-aware attention mechanism that explicitly queries matching pixels across views, enabling holistic panorama generation from a set of overlapping perspective images. Wonder3D [96] takes a two-domain approach: a cross-domain diffusion model simultaneously denoises an RGB and a surface-normal image, with the normal output serving as high-quality geometric supervision for downstream mesh reconstruction.

Scalable and pose-free generation. EscherNet [97] generalises pose conditioning via a novel *camera positional encoding* (CaPE), enabling generation of an arbitrary number of views at any resolution without per-angle fine-tuning. ViVid-1-to-3 [169] recasts NVS as a video generation problem: the temporal axis of a pre-trained video diffusion model is repurposed to model the viewpoint axis, using the model's temporal consistency as a proxy for multi-view coherence. GeoNVS [168] addresses geometric fidelity by conditioning a video diffusion model on a rendered point cloud at the target pose, grounding the generative hallucination in explicit 3-D geometry.

7.2.2 Multi-View Diffusion and 3-D-Consistent Generation

Single-view methods generate views independently or with limited cross-view coupling. A more ambitious goal is to train a diffusion model that *natively* produces multi-view consistent outputs—effectively encoding 3-D structure implicitly in the model weights.

MVDream [101] achieves this by replacing the temporal self-attention in a video diffusion transformer with a 3-D-aware *multi-view attention* block that attends across all views simultaneously. Training on rendered 3-D assets (Objaverse) with known camera poses teaches the model to share structural information across views. The resulting model can be used both as a standalone multi-view generator and as a geometry-consistent score function for SDS-based text-to-3D.

GeNVS [100] takes a hybrid approach: a NeRF encoder constructs a 3-D feature volume from the conditioning images, and the diffusion decoder renders a target view by volumetrically querying this feature space. This architecture provides hard geometric grounding while preserving the expressiveness of the diffusion prior for high-frequency appearance details.

7.2.3 Camera Trajectory Conditioning and Video Diffusion NVS

A particularly important frontier is conditioning diffusion models on *continuous camera trajectories*, enabling the generation of smooth, temporally coherent novel-view sequences—the direct bridge to world modeling.

Camera trajectory as a control signal. ViewCrafter [110] conditions a video diffusion model (CogVideoX) on point-cloud renderings along a target camera path, using the rendered point cloud as a structural scaffold that prevents temporal drift while the diffusion model fills in appearance and fine geometry. Free3D [99] injects camera pose into each frame's cross-attention query, enabling a single video diffusion model to follow arbitrary camera trajectories from a single reference image. Novel View Extrapolation [172] specifically targets *large-angle extrapolation*—the regime where geometric hallucination is most necessary—using video diffusion priors to maintain perceptual plausibility when the reference content provides little direct evidence.

Training-free and zero-shot camera control. NVS-Solver [193] demonstrates that an off-the-shelf video diffusion model already encodes sufficient geometric knowledge to solve NVS in a zero-shot manner: by formulating view synthesis as a constrained diffusion process with known-region guidance, it achieves competitive performance without any pose-specific fine-tuning. DDIM Inversion for NVS [163] exploits the deterministic reconstruction property of DDIM [208]: inverting

the reference image to its latent code and re-decoding with a modified pose conditioning produces structurally faithful novel views, again without additional training.

Geometric conditioning for high fidelity. Projected Representation Conditioning [166] proposes projecting 3-D features from the reference view onto the target-view image plane, providing an explicit geometric "warp" prior to the diffusion decoder. This projected feature conditioning substantially reduces hallucination artifacts in regions visible from the target but occluded in the reference. A Single Image and Multimodality [167] augments pose conditioning with depth, surface normal, and semantic maps, each derived from monocular estimation, to enrich the conditioning signal across multiple complementary geometric cues.

7.2.4 Reconstruction–Diffusion Hybrids

A complementary design philosophy *separates* the diffusion and reconstruction stages: diffusion fills in missing views and hallucinations, while a deterministic reconstruction pipeline (NeRF or 3DGS) enforces 3-D consistency.

Cat3D [108] exemplifies this pattern: given sparse reference views ($n \geq 1$), a multi-view diffusion model generates a dense set of synthetic pseudo-views, which are then fed to a standard 3DGS reconstruction pipeline. Because the diffusion model is fine-tuned jointly with the reconstruction loss, the generated views are explicitly optimised for downstream reconstruction fidelity. ReconFusion [109] integrates the diffusion prior more tightly: it conditions a pixel-diffusion model on pixelNeRF [30] feature queries, acting as a learned regulariser that fills in uncertain regions during sparse-view NeRF optimisation. ZeroNVS [98] handles the challenging case of in-the-wild single images (non-object-centric, background present) by training on scene-level data and conditioning on the camera intrinsics alongside pose, enabling 360° generation from a single casual photograph. NVComposer [170] addresses the case where multiple reference views are available but unposed: a feature composition network aggregates information from all reference images before passing to the diffusion model, improving quality without requiring explicit pose estimation.

7.3 Score Distillation: Diffusion as a 3-D Optimisation Oracle

Score Distillation Sampling (SDS) [102] opened a separate pathway: instead of directly generating novel views, use a pre-trained 2-D diffusion model as a differentiable *score function* to optimise a 3-D representation. The core identity is:

$$\nabla_\theta \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\boldsymbol{\epsilon}} \left[w(t)(\hat{\boldsymbol{\epsilon}}_\phi(\mathbf{x}_t; y, t) - \boldsymbol{\epsilon}) \, \frac{\partial \mathbf{x}}{\partial \theta} \right], \tag{5}$$

where θ parameterises the 3-D representation, $\mathbf{x}$ is a rendered image, y is a text prompt, and $\hat{\boldsymbol{\epsilon}}_\phi$ is the diffusion model's noise prediction.

DreamFusion [102] first demonstrated that SDS can synthesise coherent 3-D objects from text descriptions. Magic3D [103] introduces a coarse-to-fine strategy that first optimises a coarse NeRF then refines a differentiable mesh, achieving $8\times$ higher resolution. ProlificDreamer [104] identifies the "over-saturation" artifact of SDS as a consequence of mode-seeking behaviour, proposing *Variational Score Distillation* (VSD) which models the 3-D parameter distribution as a particle set and applies Stein variational gradient descent for diverse, high-quality generation. Fantasia3D [105] disentangles geometry (optimised via SDS on a DMTet mesh) from appearance (a physically-based BRDF), enabling photorealistic materials and independent editing of shape and texture. GaussianDreamer [106] replaces NeRF with 3DGS as the SDS target, leveraging Gaussian densification for fine detail and achieving real-time rendering of the final asset. LucidDreamer [107] addresses SDS instability with an interval-based score matching scheme that reduces gradient variance.

7.4 Dedicated Multi-View Diffusion Architectures

The methods above predominantly adapt existing video or image diffusion models. A complementary line of work designs architectures *from scratch* for multi-view consistency.

MVDream [101] replaces temporal attention with multi-view attention and trains on Objaverse with ground-truth camera poses, producing natively 3-D-consistent outputs. ImageDream [213] extends

Table 4: Generative NVS methods. **Backbone**: LDM=latent diffusion, VDM=video diffusion, GAN=generative adversarial. **Ctrl**: pose conditioning mechanism. **3D Out**: whether the method produces an explicit 3-D representation.

Method	Venue	Backbone	Ctrl	3D Cons.	3D Out	Key Contribution
GAN-based						
GRAF [85]	NeurIPS'20	NeRF-GAN	Latent	∼	NeRF	Radiance field GAN
pi-GAN [86]	CVPR'21	NeRF-GAN	Latent	∼	NeRF	Periodic implicit
GIRAFFE [87]	CVPR'21	Feat. Field	Latent	∼	Neural	Compositional
EG3D [88]	CVPR'22	Triplane	Latent	✓	Tri.	Triplane+SR
StyleNeRF [89]	ICLR'22	StyleGAN	Latent	∼	NeRF	Style-based NeRF
Pose-conditioned single-image diffusion						
Zero-1-to-3 [91]	ICCV'23	LDM	Δpose	∼	–	Pose-cond LDM
Zero123++ [94]	arXiv'23	LDM	Δpose	∼	–	6-view grid
One-2-3-45 [93]	NeurIPS'23	LDM+MVS	Δpose	✓	Mesh	Diff→MVS
SyncDreamer [92]	ICLR'24	LDM	Δpose	✓	–	Volume sync
Wonder3D [96]	CVPR'24	LDM	Δpose	✓	–	RGB+Normal
EscherNet [97]	CVPR'24	LDM	CaPE	✓	–	Arbitrary #views
ViVid-1-to-3 [169]	arXiv'24	VDM	Δpose	∼	–	Video=NVS axis
Camera trajectory / video diffusion						
Free3D [99]	ICLR'24	VDM	Cam embed	∼	–	Per-frame pose
ViewCrafter [110]	arXiv'24	VDM	Ptcloud	✓	–	Scaffold+VDM
NVS-Solver [193]	arXiv'24	VDM	Zero-shot	∼	–	Guided diffusion
DDIM Inversion [163]	arXiv'24	LDM	Inversion	∼	–	Training-free
GeoNVS [168]	arXiv'24	VDM	Ptcloud	✓	–	Geo-grounded VDM
Proj. Repr. Cond. [166]	arXiv'24	LDM	3D proj.	✓	–	Feature warp
Novel View Extrap. [172]	arXiv'24	VDM	Cam traj.	∼	–	Large-angle extrap.
Reconstruction–diffusion hybrids						
GeNVS [100]	ICLR'23	LDM	NeRF feat	✓	NeRF	NeRF-cond diff.
ZeroNVS [98]	CVPR'24	LDM	Pose+int.	∼	–	In-the-wild
ReconFusion [109]	CVPR'24	LDM	pixelNeRF	✓	NeRF	Diff reg.
Cat3D [108]	NeurIPS'24	MV-Diff	Δpose	✓	3DGS	MV gen+recon
NVComposer [170]	arXiv'24	LDM	Feature	✓	–	Unposed multi-ref
Score distillation (text-to-3D)						
DreamFusion [102]	ICLR'23	LDM+NeRF	SDS	∼	NeRF	SDS
Magic3D [103]	CVPR'23	LDM+Mesh	SDS	∼	Mesh	C2F SDS
ProlificDreamer [104]	NeurIPS'23	LDM+NeRF	VSD	∼	NeRF	VSD
Fantasia3D [105]	ICCV'23	LDM+Mesh	SDS	∼	BRDF	Disentangled
GaussianDreamer [106]	CVPR'24	LDM+3DGS	SDS	∼	3DGS	SDS+3DGS

MVDream to image-conditioned generation, enabling single-image-to-3D with strong semantic fidelity. Zero123-XL [212] scales the Zero-1-to-3 paradigm to larger model capacities and more diverse training data. Direct3D [214] trains a native 3-D latent diffusion model on a triplane latent space, enabling direct 3-D generation without a two-stage pipeline.

Table 4 provides a structured comparison of generative NVS methods across the four paradigms above.

Implications for World Modeling. Diffusion-based NVS represents a critical advance along World Modeling Axis 2 (sparse/single-image generalisation, **D2**) and provides the generative prior essential for completing unobserved scene regions (Axis 1, geometric consistency **D1**). Camera trajectory conditioning (§7.2) directly bridges into Axis 3 (temporal dynamics), as a camera trajectory can be viewed as a special case of a time-ordered action sequence. The remaining gap is *physical action conditioning* (**D4, D5**): existing diffusion NVS methods condition on camera parameters but not on object-level physical interactions, which remains an open problem addressed in §11.

8 Towards Generative World Modeling

8.1 Dynamic Scenes and 4-D World Representation

The step from static NVS to world modeling hinges on modelling *temporal dynamics*: how does a scene evolve over time, and can the model generate novel views at arbitrary future timestamps? The dynamic NeRF and 4DGS methods in §4–5 address this for *observed* scenes given dense multi-view video. A harder and more practically relevant regime is *generalizable* 4-D generation from sparse or single-image input—precisely the regime inhabited by world models.

Video diffusion models have emerged as the most promising backbone. ViewCrafter [110] augments a video diffusion model with point-cloud renderings along the target camera trajectory, providing a geometric scaffold that constrains the generation while the diffusion prior fills in appearance and fine detail. CAT4D [199] generalises the Cat3D framework to the temporal dimension, jointly modelling viewpoint and time with a multi-view video diffusion model, then lifting the output to an explicit 4-D representation. 4DiM [111] takes a more direct approach, training a 4-D diffusion model that simultaneously denoises a set of frames at varying timestamps and viewpoints, learning a joint spatiotemporal prior. SV3D [198] and SV4D [112] extend Stable Video Diffusion to 3-D and 4-D object generation respectively, conditioning on orbital camera paths to ensure multi-view consistency. Consistent4D [113] enforces temporal consistency across a 4-D cascade, resolving inter-frame Gaussian drift. DreamGaussian4D [114] shows that SDS can be applied to 4DGS, enabling text-driven 4-D generation without any video training data.

A complementary line of work addresses the *monocular* case. Diffusion Priors for Dynamic NVS [180] treats the diffusion model as a frame-level regulariser during per-scene 4-D optimisation from a single monocular video, substantially improving novel-view quality in under-constrained regions.

8.2 Neural Scene Simulators for Autonomous Driving

Autonomous driving provides a uniquely demanding test-bed for NVS-as-world-model: scenes are large-scale, unbounded, and contain independently moving actors (vehicles, pedestrians) that must be rendered consistently from closed-loop ego-motion trajectories. Three architectural strategies have emerged.

NeRF-based compositional simulators. MARS [119] separates the scene into a static NeRF background and per-agent NeRF foregrounds, allowing independent actor manipulation. UniSim [118] extends this to a closed-loop sensor simulator that supports controllable scene editing—removing actors, changing weather, inserting novel agents—with photorealistic rendering at the ego sensor. EmerNeRF [121] learns the dynamic-static decomposition in a self-supervised manner by decomposing scene flow from multi-camera video, without per-actor instance labels.

3DGS-based driving reconstruction. Street Gaussians [122] extends 3DGS to urban scenes by representing tracked foreground objects with separate Gaussian models and compositing them over a background Gaussian field. PVG [123] models Gaussian motion with periodic vibration functions that efficiently encode slow, cyclic motions common in urban traffic. OmniRe [120] unifies background, foreground, and sky in a single Gaussian framework, establishing a strong all-in-one baseline.

Diffusion-based street generation. StreetCrafter [200] uses a point-cloud-conditioned video diffusion model with controllable camera motion, enabling photorealistic street-view synthesis for data augmentation. Sat2Vid [190] lifts a satellite image to a street-view panoramic video, addressing the altitude gap between aerial mapping and ground-level perception. Wild-GS [171] solves the unconstrained internet-photo case by disentangling transient and persistent Gaussian appearance in 3DGS. InSpatio-WorldFM [203] is an open-source real-time generative frame model targeting spatial world understanding at scale.

Table 5: World modeling and neural scene simulation methods.

Method	Year	3D Aware	Video	Action	Domain	Key Feature
GAIA-1 [117]	2023	∼	✓	✓	Driving	Seq. token WM
UniSim [118]	2023	✓	✓	✓	Driving	Sensor sim.
MARS [119]	2023	✓	✓	∼	Driving	Compositional
Genie [124]	2024	–	✓	✓	Games	Latent action
Genie 2 [125]	2024	✓	✓	✓	Open-world	Interactive 3D
WonderWorld [129]	2024	✓	–	–	Open-world	Instant 3DGS
ViewCrafter [110]	2024	✓	✓	–	Open-world	Cam-cond video
Cosmos [126]	2025	✓	✓	✓	Physical	WFM training
Street Gaussians [122]	2024	✓	✓	∼	Driving	3DGS driving
OmniRe [120]	2024	✓	✓	✓	Driving	All-in-one

8.3 Generative World Models and Foundation Models

World foundation models represent the apex of the NVS-to-world-modeling trajectory: systems that jointly handle viewpoint, time, and action, trained at internet scale. We identify three design axes that differentiate current approaches.

Sequence vs. spatial representation. GAIA-1 [117] treats driving scenes as sequences of video, text, and action tokens, training an autoregressive transformer to predict the next token. This approach achieves strong temporal coherence but lacks an explicit 3-D representation, limiting viewpoint flexibility. Genie [124] learns a latent action space from unsupervised internet videos, enabling interactive game-world generation without action labels—an important step toward self-supervised world model pretraining.

3-D-grounded generation. Genie 2 [125] introduces persistent 3-D structure into the Genie framework, generating scenes that maintain geometry across viewpoints and support first-person interactive navigation—the closest current system to an NVS-grounded world model for open-ended environments. WonderWorld [129] takes a complementary approach: instead of learning from large-scale video, it uses instant 3DGS initialisation from a single image combined with diffusion inpainting to generate interactive 3-D scenes on the fly. WonderJourney [128] chains text-guided scene generation steps to produce perpetual explorable 3-D environments.

Physical-world foundation models. Cosmos [126] is the most ambitious current effort: a family of world foundation models pretrained on internet-scale video with explicit physical world conditioning, designed for downstream fine-tuning to robotics and autonomous driving. Pandora [127] interleaves language and video tokens, enabling language-driven world simulation with both visual and linguistic output.

Table 5 compares world modeling methods.

8.4 Embodied AI and Robotics

NVS is more than a rendering tool in robotics: it provides the *spatial prior* that allows agents to reason about unseen parts of the environment before taking action. Three use patterns have emerged.

Pose estimation and object localisation. iNeRF [131] inverts a trained NeRF to estimate 6-DoF camera pose by gradient descent on the rendered photometric error—the first demonstration that NeRF's differentiable renderer can be repurposed for inference as well as reconstruction. Dex-NeRF [130] applies NeRF to transparent-object grasping, where conventional depth sensors fail, by leveraging NeRF's ability to model view-dependent appearance to infer geometry from specular reflections.

Visual planning and manipulation. NeRF-Supervisor [132] uses a pre-trained NeRF to synthesise robot-wrist camera views at hypothetical future poses, enabling visual planning without physical rollouts. GaussianGrasping [133] replaces NeRF with 3DGS for real-time manipulation planning, exploiting the explicit Gaussian representation for fast collision checking and grasp pose optimisation.

Table 6: NVS vs. World Modeling capability boundary.

Capability	NVS (static)	World Model
Viewpoint-consistent rendering (D1)	✓	✓
Sparse/single-image input (D2)	∼	✓
Temporal dynamics (D3)	–	✓
Action/agent conditioning (D4)	–	✓
Physical plausibility (D5)	–	✓
Cross-scene generalisation	∼	✓
Causal scene understanding	–	✓

Simulation-to-real transfer. RoboGSim [134] reconstructs robot workspaces in 3DGS and renders synthetic training demonstrations from novel viewpoints, directly closing the sim-to-real gap for visuomotor policy learning.

8.5 Human Novel View Synthesis

Synthesising novel views of humans is a critical subproblem with applications in AR/VR, telepresence, and embodied AI.

Neural implicit human models. Neural Body [189] anchors structured latent codes on a parametric human body (SMPL), enabling novel view and pose synthesis. Rotationally-Temporally Consistent [188] enforces both rotation and temporal consistency for human performance video synthesis. NHR [150] adopts multi-view point-cloud-based rendering for dynamic human synthesis.

Generalizable and diffusion-based human NVS. CFSynthesis [178] enables controllable and free-view 3-D human video synthesis via a disentangled representation. iVS-Net [181] learns human view synthesis from internet videos without any 3-D supervision. Blur2Sharp [187] addresses human novel pose and view synthesis from blurry input with a generative prior. Diffuman4D [186] achieves 4-D consistent human view synthesis from sparse-view videos via video diffusion. EVA-Gaussian [185] achieves real-time 3DGS-based human NVS under diverse multi-view settings.

8.6 The Boundary: NVS vs. World Modeling

Table 6 delineates the capability boundary.

9 Datasets and Benchmarks

9.1 Object-Level Datasets

ShapeNet [145]: 51K 3-D CAD models; foundation for single-image reconstruction training. **Objaverse** [146]: 800K+ 3-D objects; enables LRM-scale training. **Objaverse-XL** [147]: 10M+ 3-D assets. **Co3D** [148]: 19K+ annotated per-category real video sequences for category-level NVS. **MVImgNet** [149]: 238K multi-view real images across 238 categories.

9.2 Static Scene Datasets

NeRF-Synthetic [17]: 8 synthetic bounded objects. **LLFF** [139]: 8 forward-facing real scenes. **DTU** [140]: 124 controlled MVS scenes. **Tanks and Temples** [141]: large-scale indoor/outdoor. **Mip-NeRF 360** [20]: 9 unbounded 360° scenes. **ScanNet** [142]: 1513 indoor RGB-D sequences. **ScanNet++** [143]: DSLR + iPhone high-resolution indoor. **RealEstate10K** [144]: 10K real estate videos for generalizable NVS training.

9.3 Dynamic Scene Datasets

D-NeRF [37]: 6 monocular synthetic dynamic objects. **HyperNeRF** [40]: monocular topological changes. **NHR** [150]: multi-view human performance. **CMU Panoptic** [151]: dense multi-view

Table 7: NVS and world modeling datasets. **Type**: RT=real, Syn=synthetic. **Dyn.**: contains dynamic content. **Depth**: provides ground-truth or LiDAR depth. **Pose**: provides camera/LiDAR extrinsics. **Sem.**: provides semantic / instance labels. **Scale**: approximate number of scenes/objects/sequences. **Primary use**: main application in the NVS literature.

Dataset	Year	Type	Dyn.	Depth	Pose	Sem.	Scale	Primary Use
Object-centric								
ShapeNet [145]	2015	Syn	–	✓	✓	✓	51K obj.	Generative 3-D training
NeRF-Synthetic [17]	2020	Syn	–	✓	✓	–	8 obj.	Per-scene NeRF evaluation
Co3Dv2 [148]	2021	RT	∼	–	✓	✓	19K seq.	Category-level NVS
Objaverse [146]	2023	Syn	∼	✓	✓	∼	800K obj.	LRM / diffusion training
Objaverse-XL [147]	2024	Syn	∼	✓	✓	∼	10M+ obj.	Large-scale 3-D pretraining
MVImgNet [149]	2023	RT	–	–	✓	✓	238K seq.	Category generalizable NVS
Static indoor / outdoor scenes								
DTU [140]	2014	RT	–	✓	✓	–	124 scn.	MVS / NVS evaluation
LLFF [139]	2019	RT	–	–	✓	–	8 scn.	Forward-facing NeRF
Tanks&Temples [141]	2017	RT	–	✓	✓	–	14 scn.	Large-scale reconstruction
Mip-NeRF360 [20]	2022	RT	–	–	✓	–	9 scn.	Unbounded 360° NVS
ScanNet [142]	2017	RT	–	✓	✓	✓	1513 seq.	Indoor 3-D reconstruction
ScanNet++ [143]	2023	RT	–	✓	✓	✓	460 scn.	Generalizable NVS, hi-res
RealEstate10K [144]	2018	RT	–	–	✓	–	10K vid.	Generalizable video NVS
Dynamic scenes								
D-NeRF [37]	2021	Syn	✓	–	✓	–	6 obj.	Dynamic NeRF evaluation
HyperNeRF [40]	2021	RT	✓	–	✓	–	15 seq.	Topological deformation
CMU Panoptic [151]	2015	RT	✓	✓	✓	✓	65 seq.	Multi-view social activities
Neu3D [41]	2022	RT	✓	–	✓	–	6 seq.	Multi-view human video NeRF
iPhone Dataset [152]	2022	RT	✓	–	–	–	14 seq.	Casually-captured dynamic NVS
Autonomous driving & large-scale								
KITTI [204]	2013	RT	✓	✓	✓	✓	Urban	Early driving benchmark
Waymo Open [153]	2020	RT	✓	✓	✓	✓	1000 seg.	Driving sensor simulation
nuScenes [154]	2020	RT	✓	∼	✓	✓	1000 scn.	360° driving world model
KITTI-360 [155]	2022	RT	✓	✓	✓	✓	Urban	Urban scene NVS + panoptic
Argoverse 2 [156]	2023	RT	✓	✓	✓	✓	1000 seq.	High-res 360° driving
PandaSet [157]	2020	RT	✓	✓	✓	✓	103 scn.	LiDAR+camera NVS

social activities. **Neu3D** [41]: multi-view dynamic human sequences. **iPhone Dataset** [152]: casually-captured phone videos.

9.4 Autonomous Driving and Large-Scale

Waymo Open [153]: multi-camera + LiDAR driving. **nuScenes** [154]: 360° driving + HD maps. **KITTI-360** [155]: urban driving with panoptic labels. **Argoverse 2** [156]: high-resolution 360° driving. **PandaSet** [157]: urban LiDAR+camera dataset.

Table 7 provides a structured comparison of key datasets.

10 Evaluation Metrics and Protocols

10.1 Standard Image Quality Metrics

PSNR (peak signal-to-noise ratio): pixel-level fidelity. **SSIM** [158]: structural similarity. **LPIPS** [159]: perceptual similarity via deep features; best correlated with human judgement.

Table 8 shows representative benchmark results on NeRF-Synthetic and LLFF.

10.2 Video and World Modelling Metrics

FVD [160] (Fréchet Video Distance): distributional quality of generated video sequences. **FID** [161]: Fréchet Inception Distance for image quality. **Temporal consistency**: average cosine similarity of DINO features between consecutive frames. **Task-specific**: for driving, closed-loop collision rate, route completion, imitation gap; for robotics, task success rate.

Table 8: Quantitative NVS comparison on three standard benchmarks. NeRF-Synthetic: bounded synthetic objects (8 scenes). LLFF: real forward-facing scenes (8 scenes). Mip-NeRF360: real unbounded 360° scenes (9 scenes). Higher PSNR/SSIM and lower LPIPS are better. "–" indicates not reported on that benchmark. † denotes generalizable methods evaluated without per-scene fine-tuning.

Method	**Venue**	**NeRF-Synthetic**			**LLFF**			**Mip-NeRF360**		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Per-scene NeRF										
NeRF [17]	ECCV'20	31.01	0.947	0.081	26.50	0.811	0.250	–	–	–
Mip-NeRF [19]	ICCV'21	33.09	0.961	0.043	26.70	0.829	0.177	–	–	–
TensoRF [24]	ECCV'22	33.14	0.963	0.047	26.73	0.839	0.204	–	–	–
Instant-NGP [23]	TOG'22	33.18	0.963	0.037	25.59	0.817	0.224	–	–	–
Mip-NeRF360 [20]	CVPR'22	–	–	–	29.40	0.901	0.127	29.40	0.840	0.204
Zip-NeRF [21]	ICCV'23	–	–	–	30.38	0.921	0.100	30.38	0.866	0.179
Per-scene 3DGS										
3DGS [46]	TOG'23	33.32	0.963	0.044	27.21	0.815	0.183	27.21	0.815	0.183
Mip-Splatting [47]	CVPR'24	34.66	0.973	0.019	27.79	0.827	0.163	27.92	0.842	0.158
Scaffold-GS [49]	CVPR'24	33.56	0.966	0.033	27.63	0.827	0.176	27.71	0.831	0.171
2DGS [55]	SIGGRAPH'24	33.35	0.965	0.042	26.94	0.818	0.212	–	–	–
Generalizable (no per-scene training)†										
pixelNeRF [30]	CVPR'21	23.17	0.860	0.170	–	–	–	–	–	–
IBRNet [31]	CVPR'21	26.73	0.917	0.091	26.73	0.851	0.176	–	–	–
MVSNeRF [32]	ICCV'21	26.63	0.931	0.168	26.40	0.839	0.181	–	–	–
pixelSplat [66]	CVPR'24	–	–	–	26.31	0.815	0.200	–	–	–
MVSplat [67]	ECCV'24	–	–	–	26.40	0.823	0.188	–	–	–

10.3 Limitations of Current Evaluation

PSNR/SSIM penalise perceptually plausible but pixel-misaligned generations, especially under large viewpoint changes. FVD captures perceptual quality but not geometric accuracy. No existing benchmark jointly evaluates geometry, dynamics, action responsiveness, and physical plausibility. This evaluation gap is itself a major open problem.

Towards systematic evaluation. The work on Systematic Evaluation of NVS [201] provides a principled analysis of how existing benchmarks differ and proposes controlled evaluation protocols separating appearance and geometry quality. Non-Aligned Reference IQA [202] addresses the fundamental problem of quality assessment when reference and synthesised views are not pixel-aligned, proposing a learned non-aligned metric for NVS. VGGT-X [165] extends VGGT to dense NVS, providing strong baselines for the emerging class of feed-forward 3-D foundation models.

11 Open Problems and Research Roadmap

11.1 The Geometry–Generation Trade-Off

Reconstruction methods deliver geometric precision but lack generative priors; diffusion methods provide rich priors but can violate multi-view consistency. Methods like ReconFusion, Cat3D, and ViewCrafter begin to bridge this gap, but the fundamental tension remains. Principled solutions (*e.g.*, geometry-grounded diffusion, diffusion-seeded 3DGS) are an active frontier.

11.2 Long-Horizon Temporal Coherence

Existing 4-D methods maintain coherence over a few seconds. Two promising directions: (1) *Persistent scene memory*: Gaussian maps that accumulate observations across time and preserve global consistency; (2) *World model pre-training*: leveraging internet-scale video prediction as a temporal prior that can be fine-tuned to specific domains.

11.3 Physical Grounding

Current methods learn appearance correlations, not physical laws. Encouraging directions include: PAC-NeRF [138] for continuum mechanics integration; physics-informed deformable Gaussians for

material simulation. The central challenge is integrating differentiable physics into the rendering pipeline in a way that scales to complex, open-world scenes.

11.4 Open-Vocabulary Semantic Integration

Methods like LERF [135] ground CLIP features in 3-D via NeRF, enabling open-vocabulary 3-D queries. LangSplat [136] brings language grounding to 3DGS. Feature3DGS [299] distils various 2-D foundation model features. Extending these to dynamic, actionable scenes is a key step toward language-grounded world models.

11.5 Evaluation and Benchmarks

We propose a five-axis world-modelling benchmark: (1) 3-D consistency under adversarial viewpoints; (2) temporal coherence over long horizons (>30 s); (3) action responsiveness in interactive settings; (4) physical plausibility via physics-engine verification; (5) open-world generalisation from a single image.

11.6 Research Roadmap

R1. Unified architectures combining explicit 3-D with diffusion priors.

R2. Internet-scale feed-forward generalisation for open-world NVS.

R3. Differentiable physics priors in 4-D world models.

R4. Language and semantic grounding for action-conditioned generation.

R5. Comprehensive world-modelling evaluation suite.

12 Conclusion

We have surveyed Novel View Synthesis through the lens of Generative World Modeling, tracing the evolution from classical image-based rendering through NeRF, 3D Gaussian Splatting, feed-forward reconstruction, and diffusion-based synthesis toward 4-D world generation. Our unified continuous-spectrum taxonomy organises more than 160 methods across three axes (input regime, temporal scope, controllability) and makes explicit their world-modelling implications.

The field is at an inflection point. Spatial grounding from NVS and temporal/action capabilities from world modeling are actively converging, as evidenced by systems such as Genie 2, Cosmos, ViewCrafter, and the neural driving simulators. Closing the geometry–generation trade-off, achieving long-horizon coherence, and grounding predictions in physical laws are the primary challenges ahead.

Draft v0.3—ongoing work. Content will be further expanded in subsequent versions.

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